POWER BI ASSIGNMENT 1
DATA TRANSFORMATION & DATA MODELING

Project Description, Key Insights and Conclusion

1. Project Description

This project focuses on preparing and integrating an e-commerce sales dataset using Power BI and Power Query Editor. The assignment uses three source tables: List of Orders, Order Details, and Sales Target. The tables contain order dates, order identifiers, customer and location details, product categories and sub-categories, quantities, amounts, profits, and monthly sales targets. The main purpose is to transform raw data into a consistent structure that can support reliable analysis and reporting.

The work shown in the screenshots includes importing the CSV files, changing data types, renaming and formatting columns, creating a Location field by combining State and City, calculating Profit Margin, classifying records as Profit, Loss, or Break-Even, merging order information with order-detail records, removing duplicate rows, sorting and filtering records, and creating relationships in the model view. These activities correspond to the assignment requirements for data preparation, data cleaning, business-rule columns, data integration, aggregation, and data modeling.

2. Work Completed Based on the Screenshots

- **Data import:** The three CSV datasets—List of Orders, Order Details, and Sales Target—are shown in Power Query Editor.
- **Data transformation:** The Order Details table has numeric data types applied to Amount, Profit, and Quantity. Customer names are formatted to proper case, and State and City are combined into a Location column.
- **Calculated columns:** A Profit Margin column is created using Profit divided by Amount, multiplied by 100. A conditional column classifies negative profit as Loss, zero profit as Break-Even, and positive profit as Profit.
- **Merging and duplicate handling:** The screenshots show Order Details expanded into a merged query and a Remove Duplicates step. A Tamil Nadu filter is also shown in the merged data.
- **Data model:** The model view shows relationships between List of Orders and Order Details, and between Order Details and Sales Target. Relationship configuration needs further validation before the model is considered complete.

3. Key Insights and Observations

- **The data is suitable for joined analysis.** Order ID is used to connect order-level information with individual order-detail rows. This allows customer and location information to be examined alongside amount, profit, quantity, category, and sub-category.
- **Profit performance varies across records.** The visible sample contains both negative and positive profit values, as well as zero-profit records. The Profit Status column is therefore useful for distinguishing losses, profitable transactions, and break-even transactions. The screenshots do not provide enough complete summary results to identify the best-performing category or total profit.
- **Profit Margin needs careful interpretation.** The formula shown is (Profit / Amount) × 100. Some records have negative margins, and a record with Amount equal to zero would need a safe handling rule to avoid a division error. Negative or unusually large percentages should be checked against the source data and business definitions.
- **Location filtering enables regional analysis.** The screenshot shows a filter for State = Tamil Nadu, with Chennai appearing as the city in the displayed records. This demonstrates how the data can be narrowed to a state for regional analysis; it does not establish statewide sales performance by itself.
- **Data-quality issues are visible in List of Orders.** The screenshots show a date-conversion step producing Error values for many rows after the first several records. The row count also shows 560 rows after a step intended to keep the first 500 rows, suggesting that the row-limiting/header steps should be reviewed. These issues should be fixed before using the table for analysis.
- **The Sales Target dates require validation.** The Sales Target preview displays dates in 2026, while the List of Orders preview shows order dates in 2018. This may reflect source data or date interpretation/locale issues. Confirm the intended month and year values and ensure the target period matches the order data before calculating target achievement.
- **The model relationships should be checked.** The model screenshot appears to show a many-to-many relationship between Order Details and Sales Target. Since both tables can contain repeated Category values, connecting them directly by Category can create ambiguous filtering or incorrect totals. A distinct Category dimension, and a proper month/date dimension where needed, would generally provide a safer model design.

4. Recommendations for Improvement

1. Promote the first row to headers immediately after importing List of Orders, then verify the column names before applying further transformations.
2. Use the correct locale when converting Order Date to Date. For dates formatted as DD-MM-YYYY, select the appropriate locale or parse the date consistently, then confirm that no Error values remain.
3. Apply Keep Top Rows = 500 to the data rows after header promotion, and verify the final row count is exactly 500 (unless the assignment interprets the limit differently).
4. Set Amount and Target to Fixed Decimal Number as requested in the questionnaire. Set Profit and Quantity to suitable numeric types as well.
5. Rename the conditional column from the default name Custom to Profit Status, and rename the merged query to Orders Data as required.
6. Check missing values and duplicates in all three tables. Remove duplicates only where the full row is genuinely duplicated; do not remove legitimate repeated Order IDs from Order Details because one order can contain multiple products.
7. Protect the Profit Margin calculation against Amount = 0 and format the resulting values as percentages for readability.
8. Validate that the Sales Target month/date values match the order-data period and use consistent month-start dates or a calendar table.
9. Review Manage Relationships. Prefer a one-to-many relationship from a unique Order ID table to Order Details, and use a distinct Category dimension to connect category-level targets. If targets are monthly, include a month/date dimension so targets filter by both category and month.
10. After correcting the data, refresh the model and test totals against the source files. Capture updated screenshots of the cleaned queries and final model view for submission.

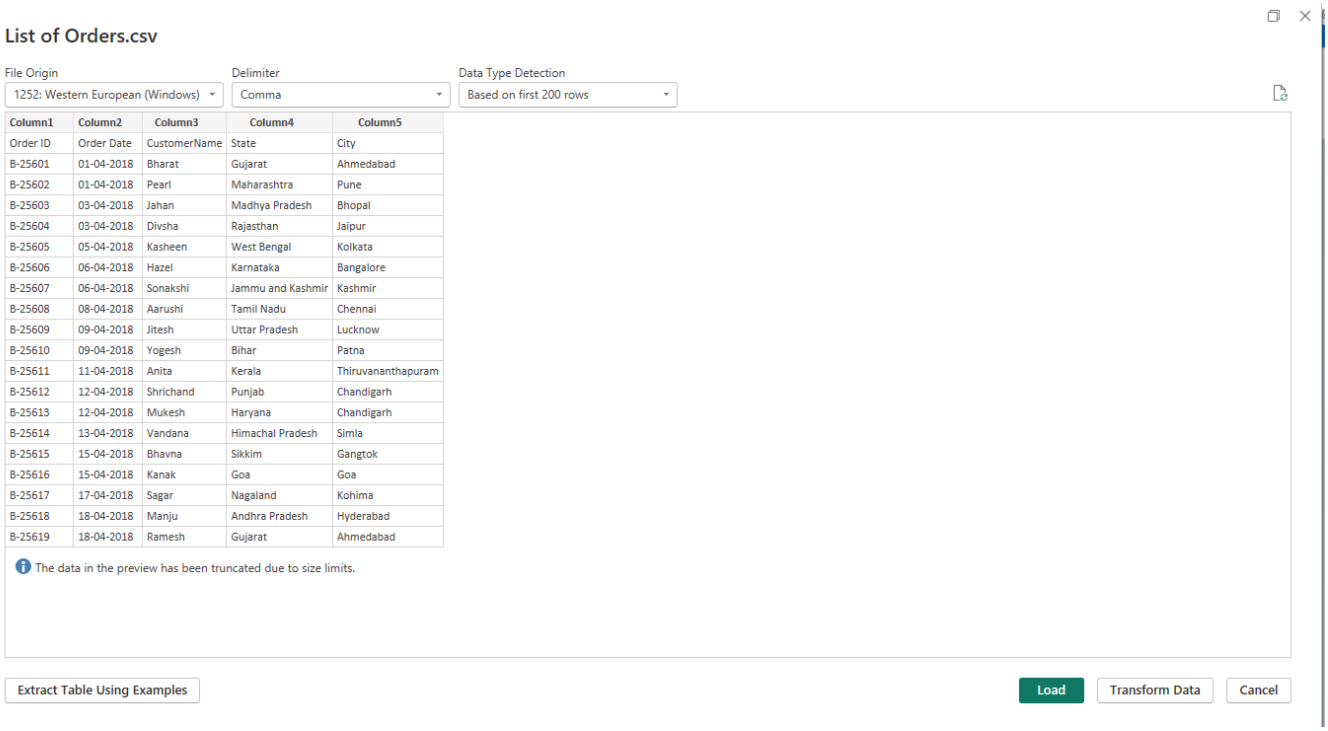
5. Conclusion

This Power BI assignment demonstrates the key stages of preparing e-commerce data for analysis, including importing multiple CSV files, transforming data types and text fields, creating location and profit-related columns, merging order and order-detail information, removing duplicates, filtering records, and establishing model relationships. The Profit Margin and Profit Status columns add useful business meaning to the transaction data, while location and category fields support further analysis by region and product group. However, the screenshots also reveal items that should be corrected before the model is treated as analysis-ready: date-conversion errors in List of Orders, a row count that does not appear to match the 500-row requirement, Sales Target dates that differ from the order-date period, and a relationship that appears to be many-to-many. Once these data-quality and modeling issues are resolved and the relationships are validated, the prepared model can provide a more reliable basis for evaluating sales, profit, product performance, and target achievement.

Screenshots of Power BI Assignment

The screenshots below are attached in the order supplied, documenting data import, Power Query transformations, calculated columns, merging, filtering, and the data model.

Screenshot 1



List of Orders.csv

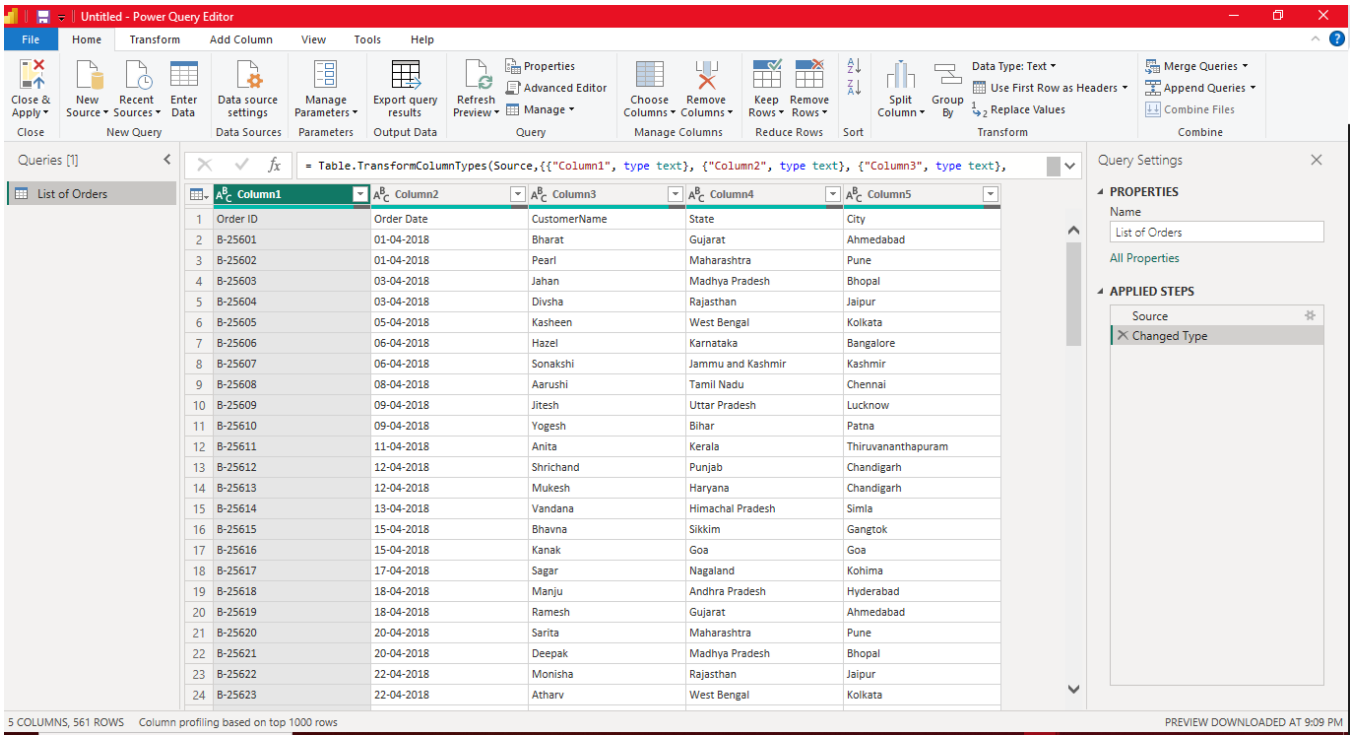
File Origin: 1252: Western European (Windows) | Delimiter: Comma | Data Type Detection: Based on first 200 rows

Column1	Column2	Column3	Column4	Column5
Order ID	Order Date	CustomerName	State	City
B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
B-25602	01-04-2018	Pearl	Maharashtra	Pune
B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
B-25606	06-04-2018	Hazel	Karnataka	Bangalore
B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
B-25610	09-04-2018	Yogesh	Bihar	Patna
B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
B-25616	15-04-2018	Kanak	Goa	Goa
B-25617	17-04-2018	Sagar	Nagaland	Kohima
B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad

The data in the preview has been truncated due to size limits.

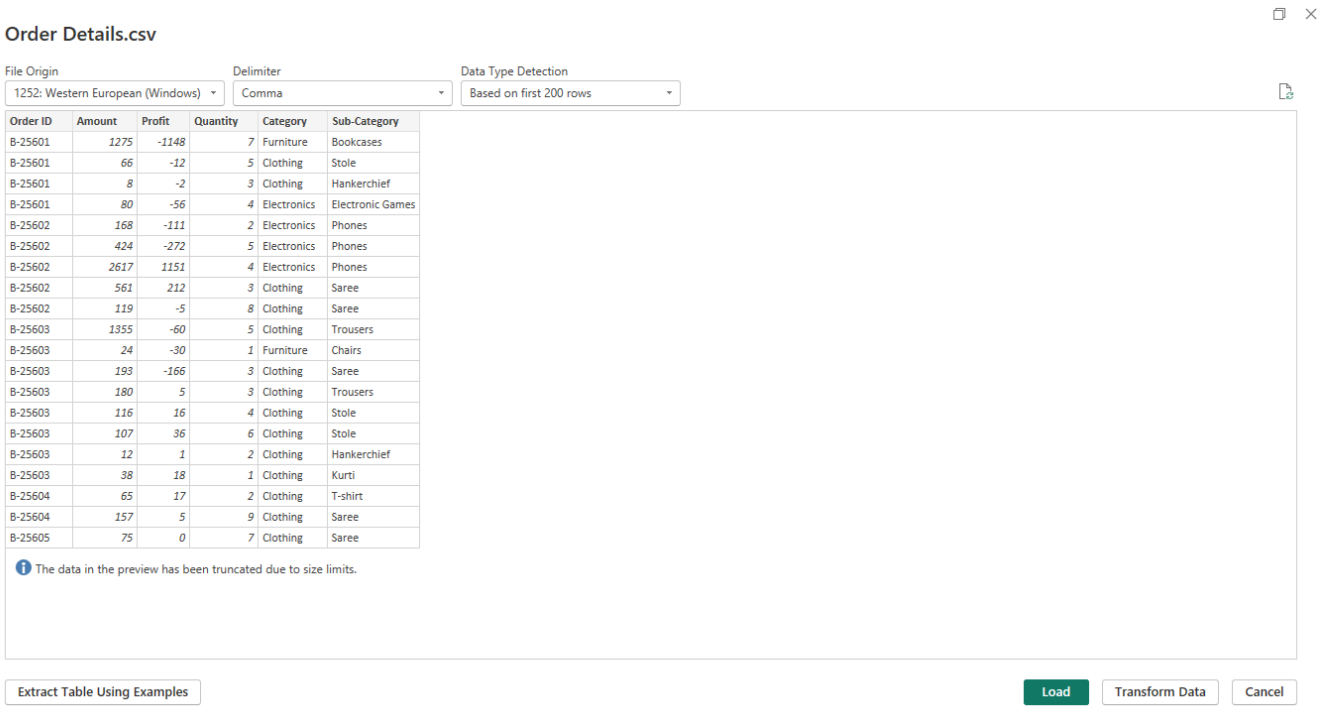
Extract Table Using Examples | Load | Transform Data | Cancel

Screenshot 2

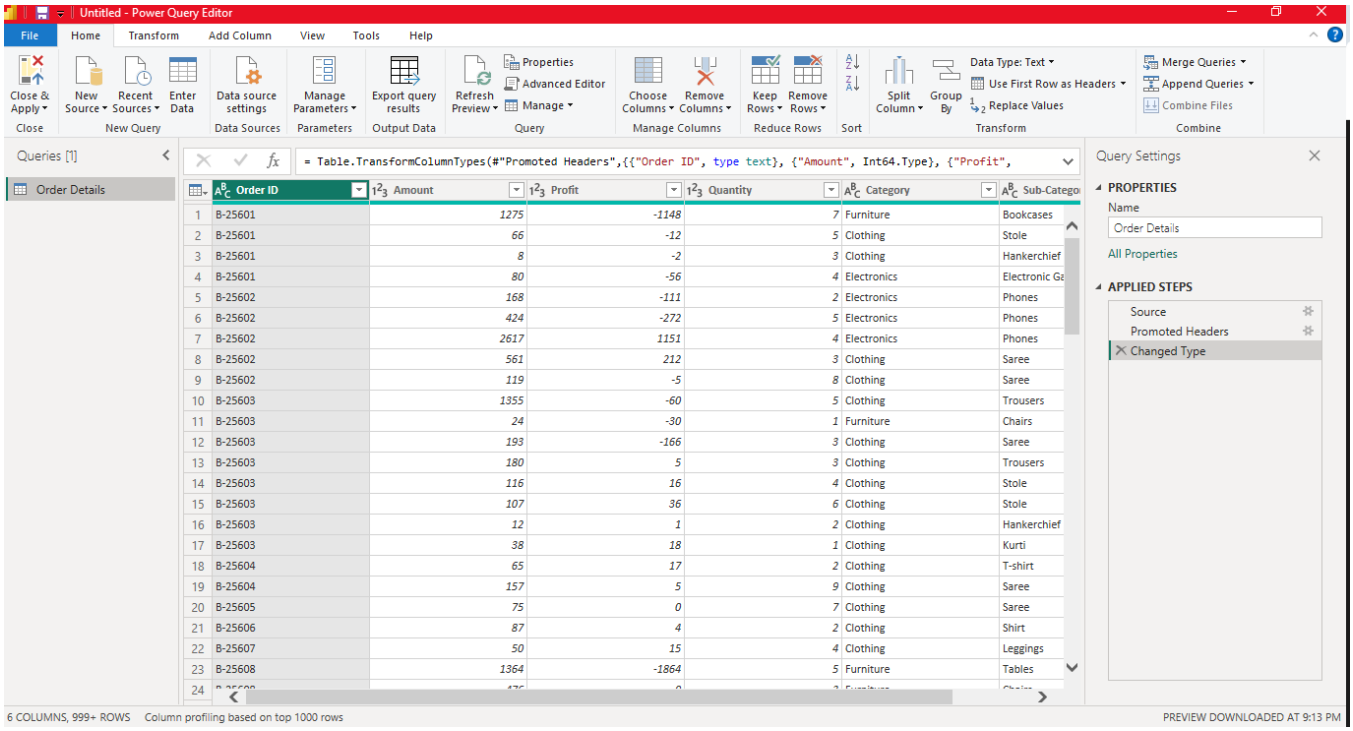


	Column1	Column2	Column3	Column4	Column5
1	Order ID	Order Date	CustomerName	State	City
2	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
3	B-25602	01-04-2018	Pearl	Maharashtra	Pune
4	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
5	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
6	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
7	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
8	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
9	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
10	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
11	B-25610	09-04-2018	Yogesh	Bihar	Patna
12	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
13	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
14	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
15	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
16	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
17	B-25616	15-04-2018	Kanak	Goa	Goa
18	B-25617	17-04-2018	Sagar	Nagaland	Kohima
19	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
20	B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
21	B-25620	20-04-2018	Sarita	Maharashtra	Pune
22	B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
23	B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
24	B-25623	22-04-2018	Atharv	West Bengal	Kolkata

Screenshot 3

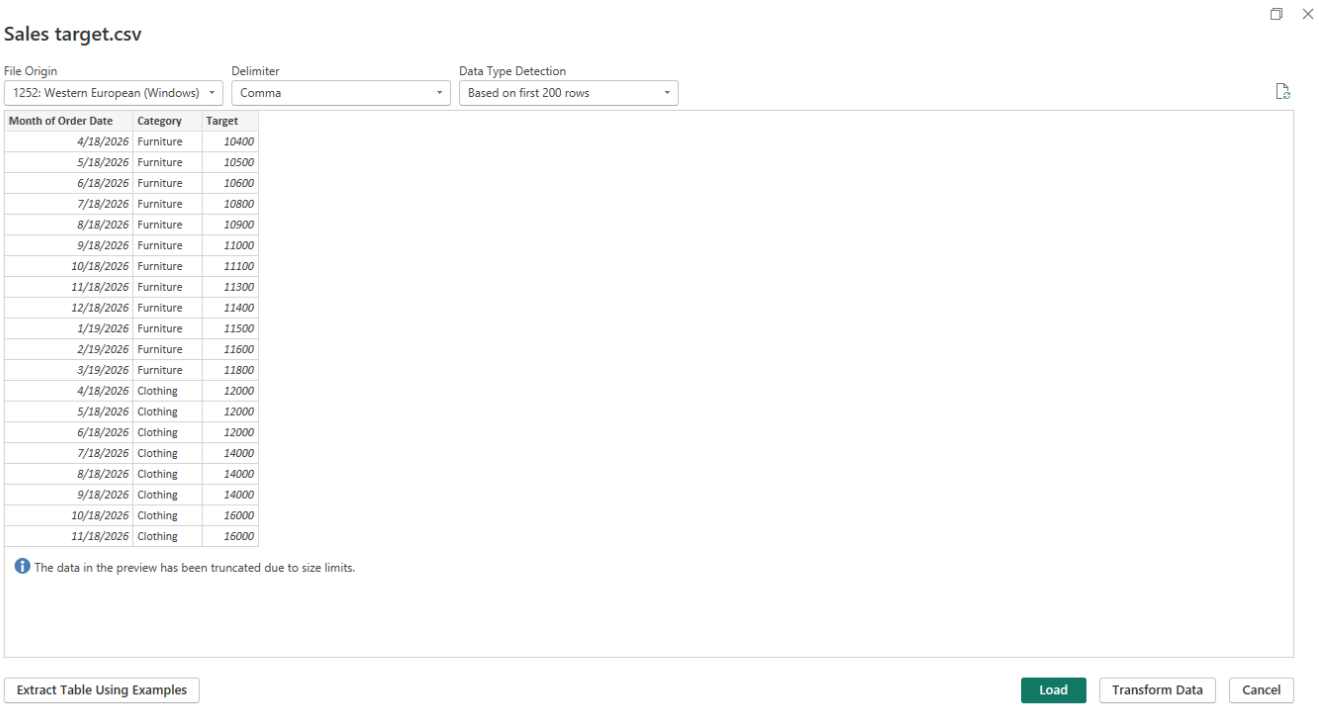


Screenshot 4

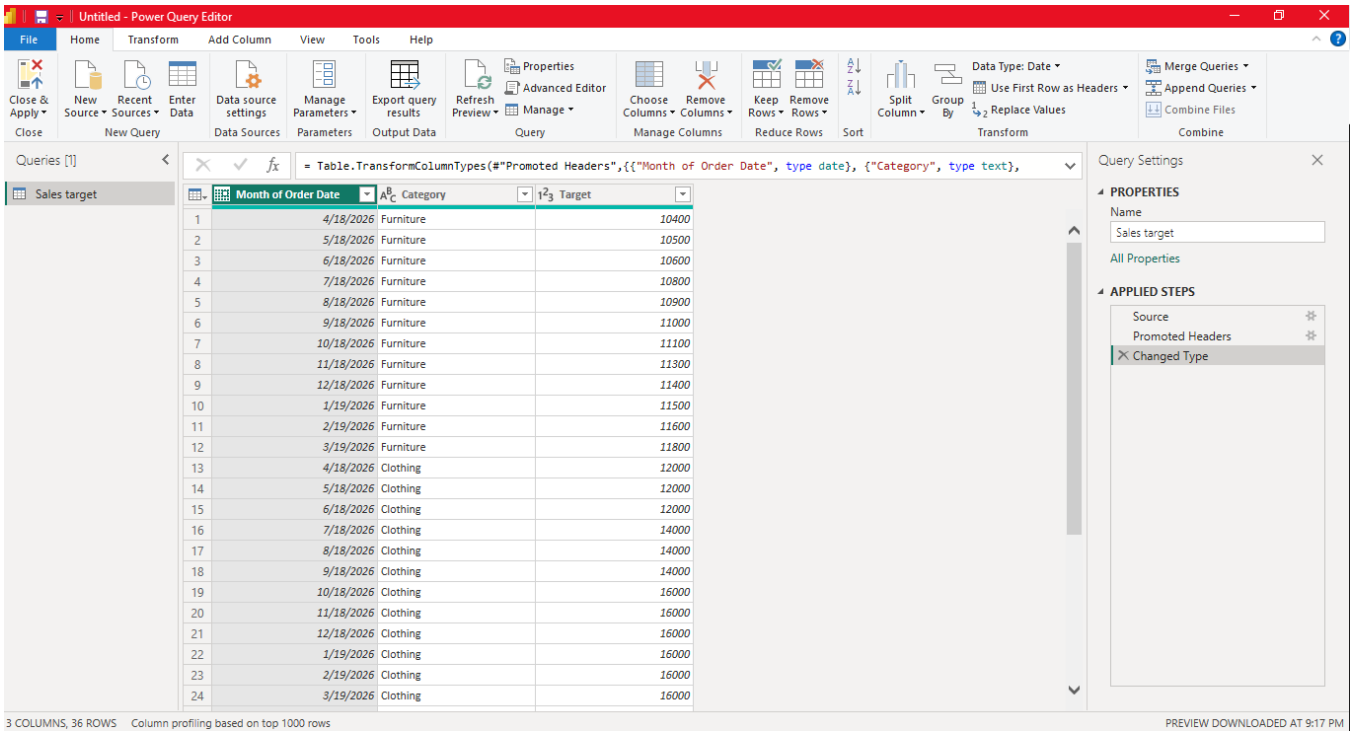


	Order ID	Amount	Profit	Quantity	Category	Sub-Categor
1	B-25601	1275	-1148	7	Furniture	Bookcases
2	B-25601	66	-12	5	Clothing	Stole
3	B-25601	8	-2	3	Clothing	Hankerchief
4	B-25601	80	-56	4	Electronics	Electronic Ga
5	B-25602	168	-111	2	Electronics	Phones
6	B-25602	424	-272	5	Electronics	Phones
7	B-25602	2617	1151	4	Electronics	Phones
8	B-25602	561	212	3	Clothing	Saree
9	B-25602	119	-5	8	Clothing	Saree
10	B-25603	1355	-60	5	Clothing	Trousers
11	B-25603	24	-30	1	Furniture	Chairs
12	B-25603	193	-166	3	Clothing	Saree
13	B-25603	180	5	3	Clothing	Trousers
14	B-25603	116	16	4	Clothing	Stole
15	B-25603	107	36	6	Clothing	Stole
16	B-25603	12	1	2	Clothing	Hankerchief
17	B-25603	38	18	1	Clothing	Kurti
18	B-25604	65	17	2	Clothing	T-shirt
19	B-25604	157	5	9	Clothing	Saree
20	B-25605	75	0	7	Clothing	Saree
21	B-25606	87	4	2	Clothing	Shirt
22	B-25607	50	15	4	Clothing	Leggings
23	B-25608	1364	-1864	5	Furniture	Tables

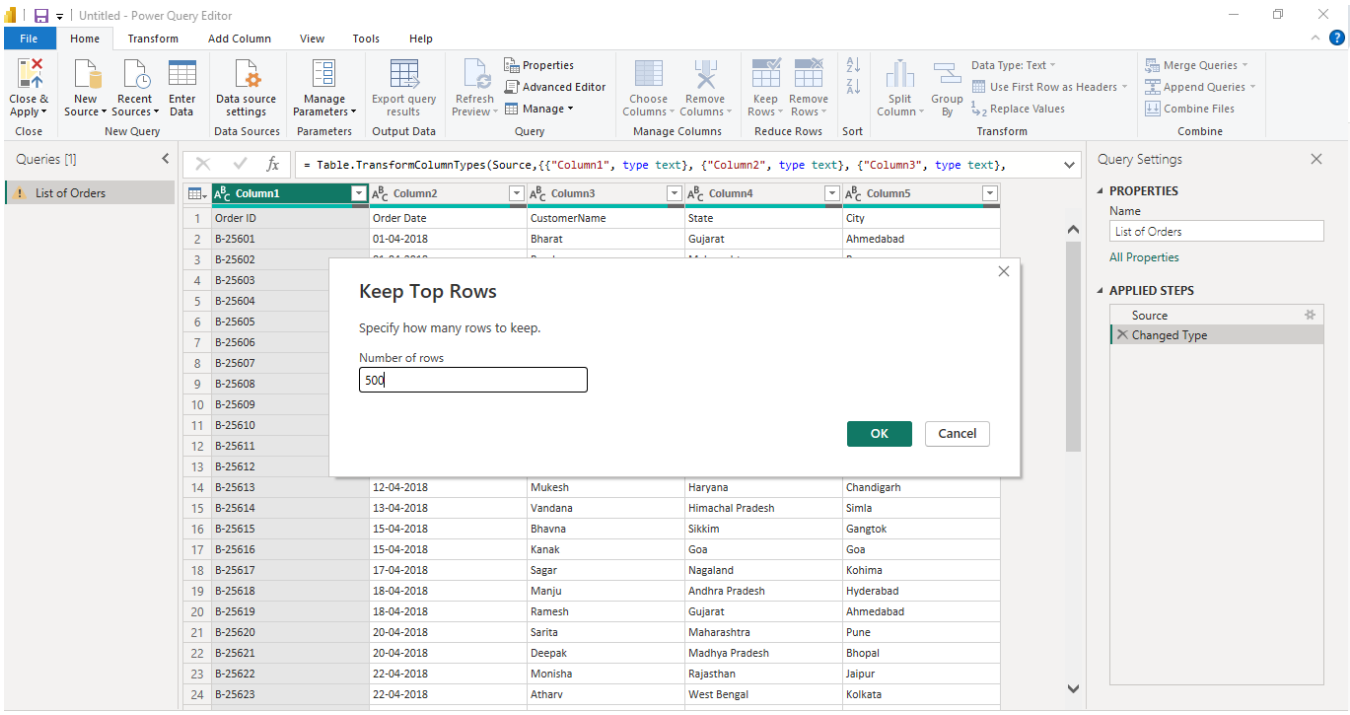
Screenshot 5



Screenshot 6

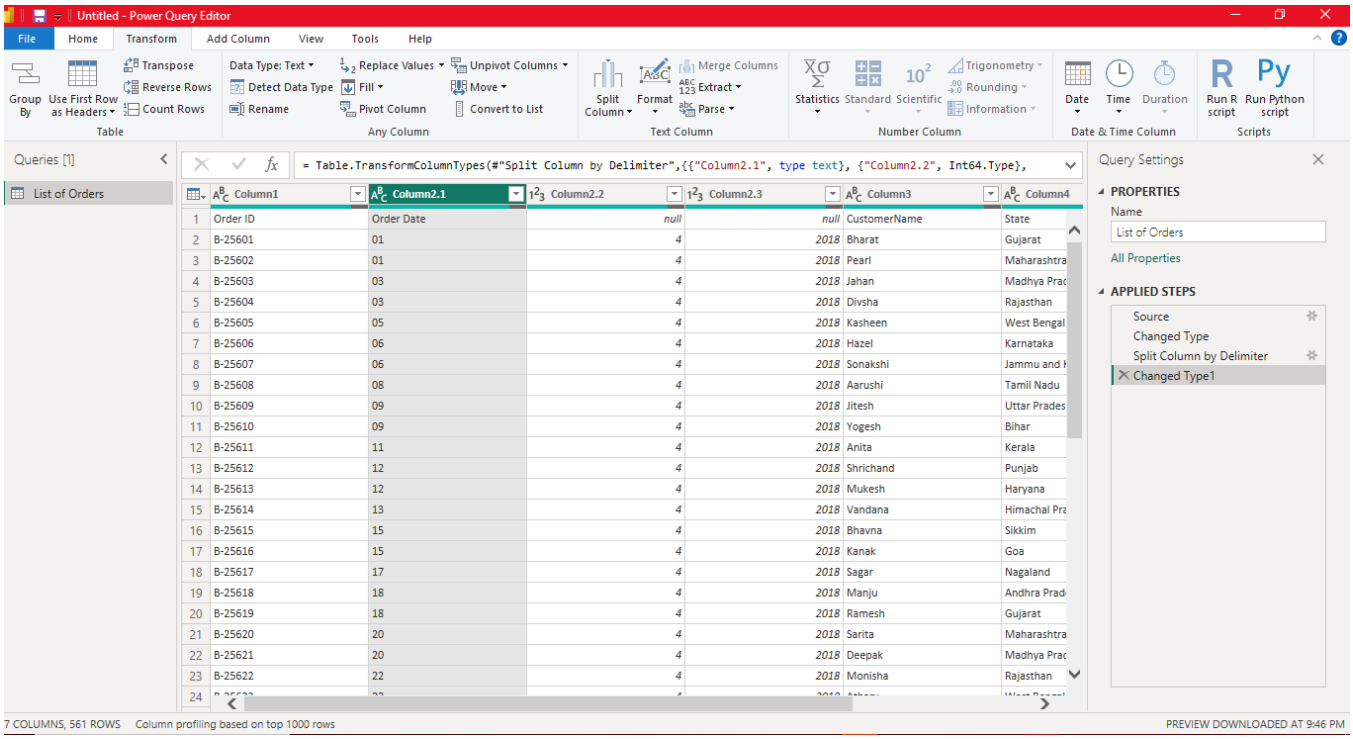


Screenshot 7

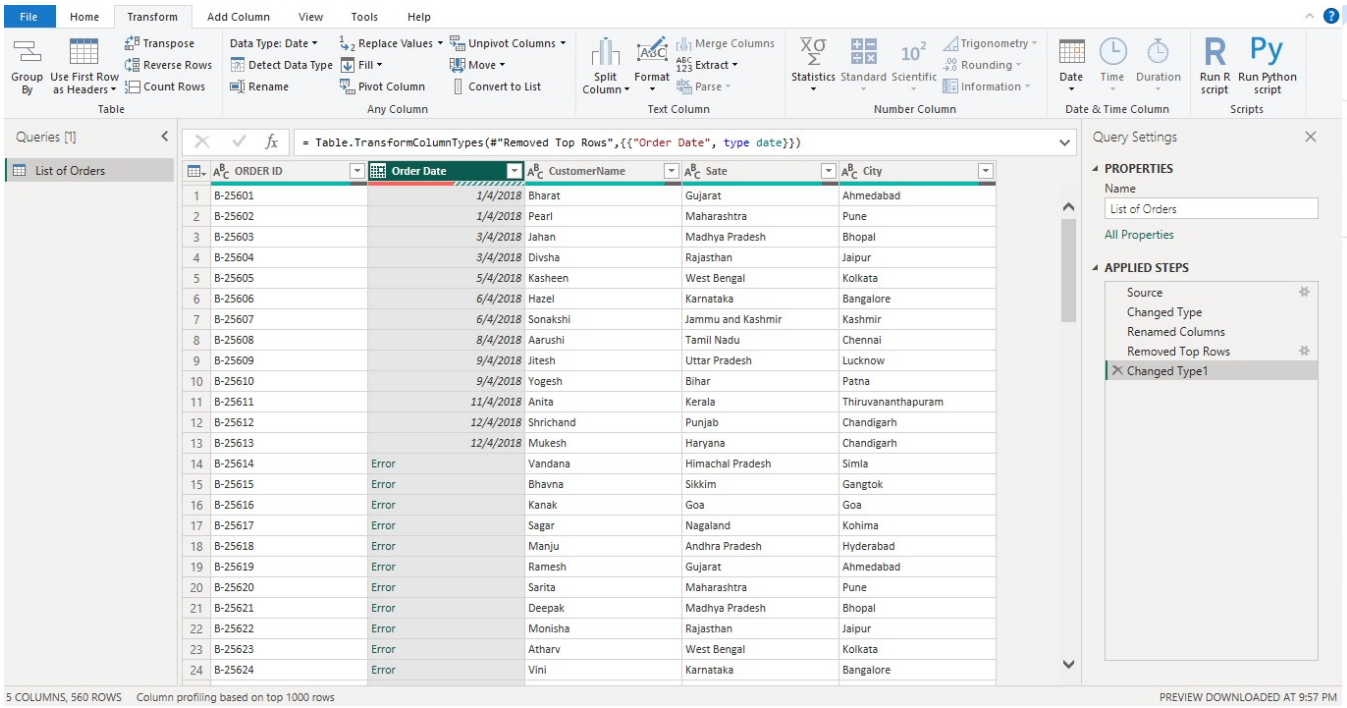


	Column1	Column2	Column3	Column4	Column5
1	Order ID	Order Date	CustomerName	State	City
2	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
3	B-25602				
4	B-25603				
5	B-25604				
6	B-25605				
7	B-25606				
8	B-25607				
9	B-25608				
10	B-25609				
11	B-25610				
12	B-25611				
13	B-25612				
14	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
15	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
16	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
17	B-25616	15-04-2018	Kanak	Goa	Goa
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23	B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
24	B-25623	22-04-2018	Atharv	West Bengal	Kolkata

Screenshot 8

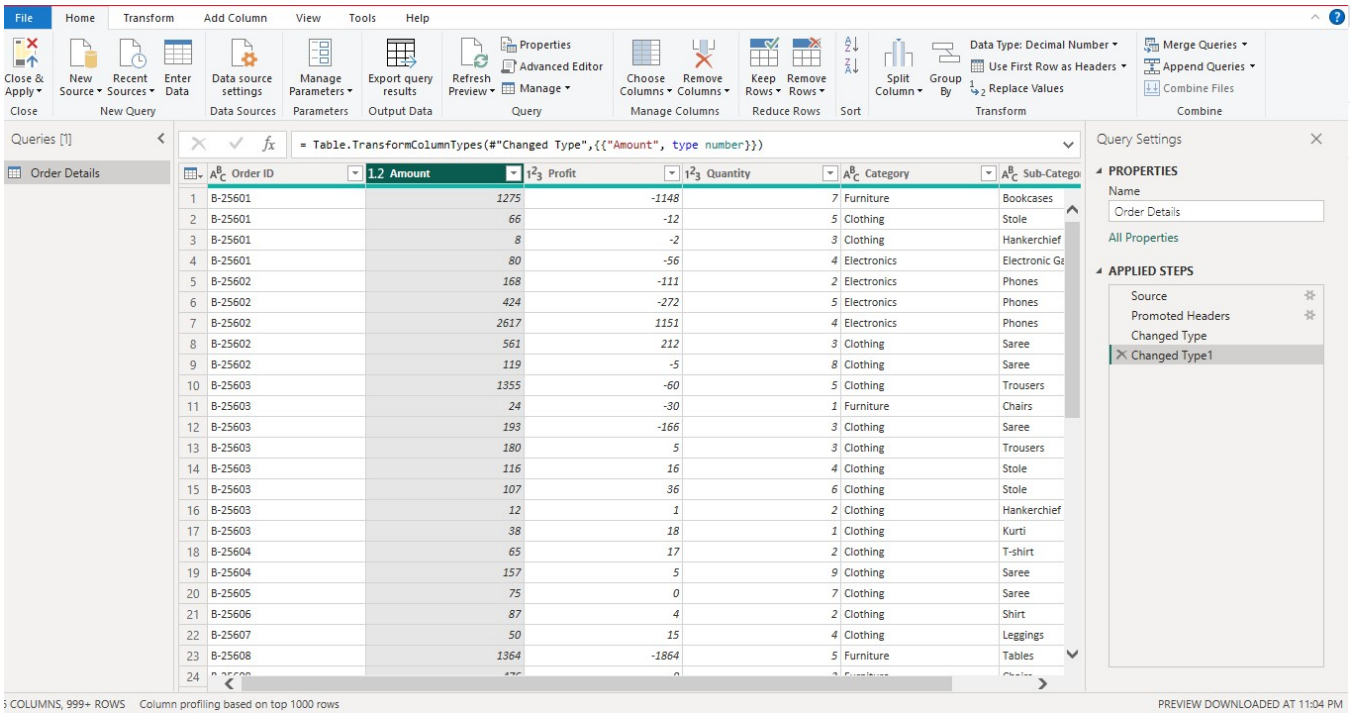


Screenshot 9

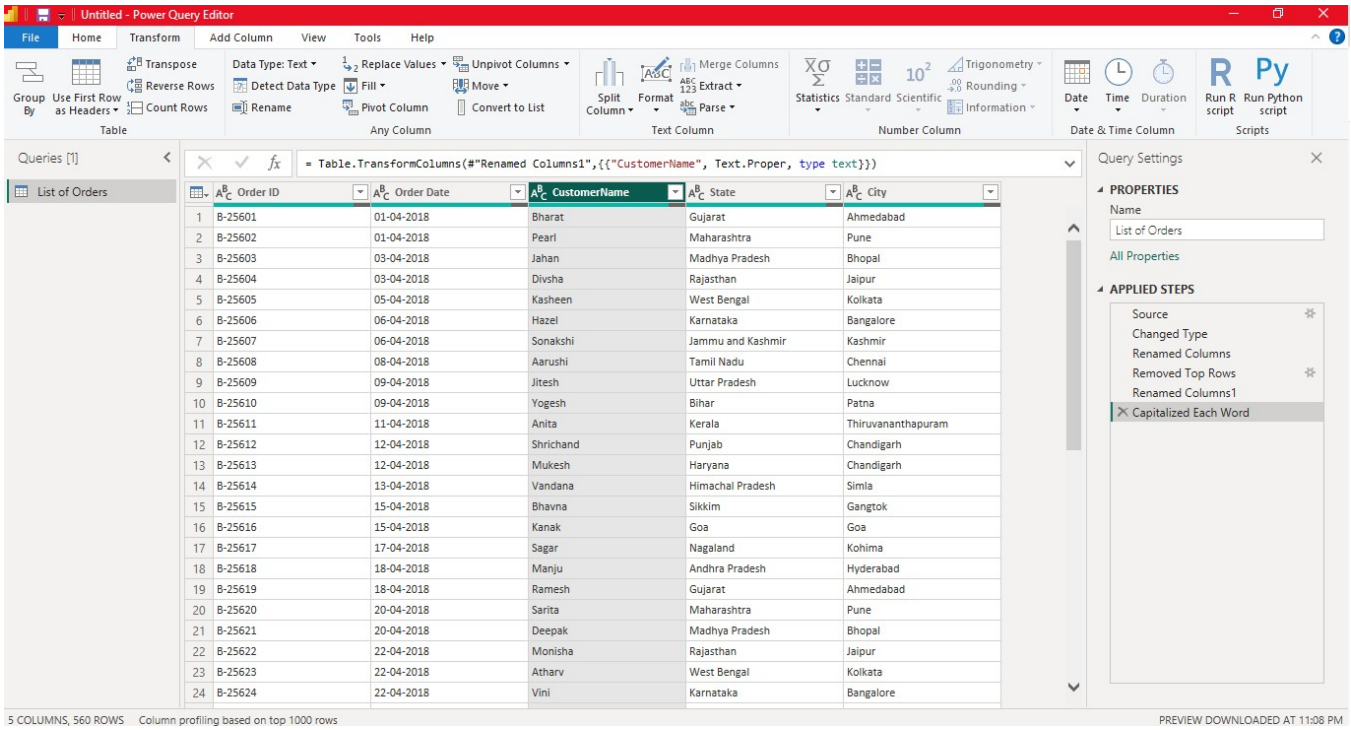


	ORDER ID	Order Date	CustomerName	Sate	City
1	B-25601	1/4/2018	Bharat	Gujarat	Ahmedabad
2	B-25602	1/4/2018	Pearl	Maharashtra	Pune
3	B-25603	3/4/2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	3/4/2018	Divsha	Rajasthan	Jaipur
5	B-25605	5/4/2018	Kasheen	West Bengal	Kolkata
6	B-25606	6/4/2018	Hazel	Karnataka	Bangalore
7	B-25607	6/4/2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	8/4/2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	9/4/2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	9/4/2018	Yogesh	Bihar	Patna
11	B-25611	11/4/2018	Anita	Kerala	Thiruvananthapuram
12	B-25612	12/4/2018	Shrichand	Punjab	Chandigarh
13	B-25613	12/4/2018	Mukesh	Haryana	Chandigarh
14	B-25614	Error	Vandana	Himachal Pradesh	Simla
15	B-25615	Error	Bhavna	Sikkim	Gangtok
16	B-25616	Error	Kanak	Goa	Goa
17	B-25617	Error	Sagar	Nagaland	Kohima
18	B-25618	Error	Manju	Andhra Pradesh	Hyderabad
19	B-25619	Error	Ramesh	Gujarat	Ahmedabad
20	B-25620	Error	Sarita	Maharashtra	Pune
21	B-25621	Error	Deepak	Madhya Pradesh	Bhopal
22	B-25622	Error	Monisha	Rajasthan	Jaipur
23	B-25623	Error	Atharv	West Bengal	Kolkata
24	B-25624	Error	Vini	Karnataka	Bangalore

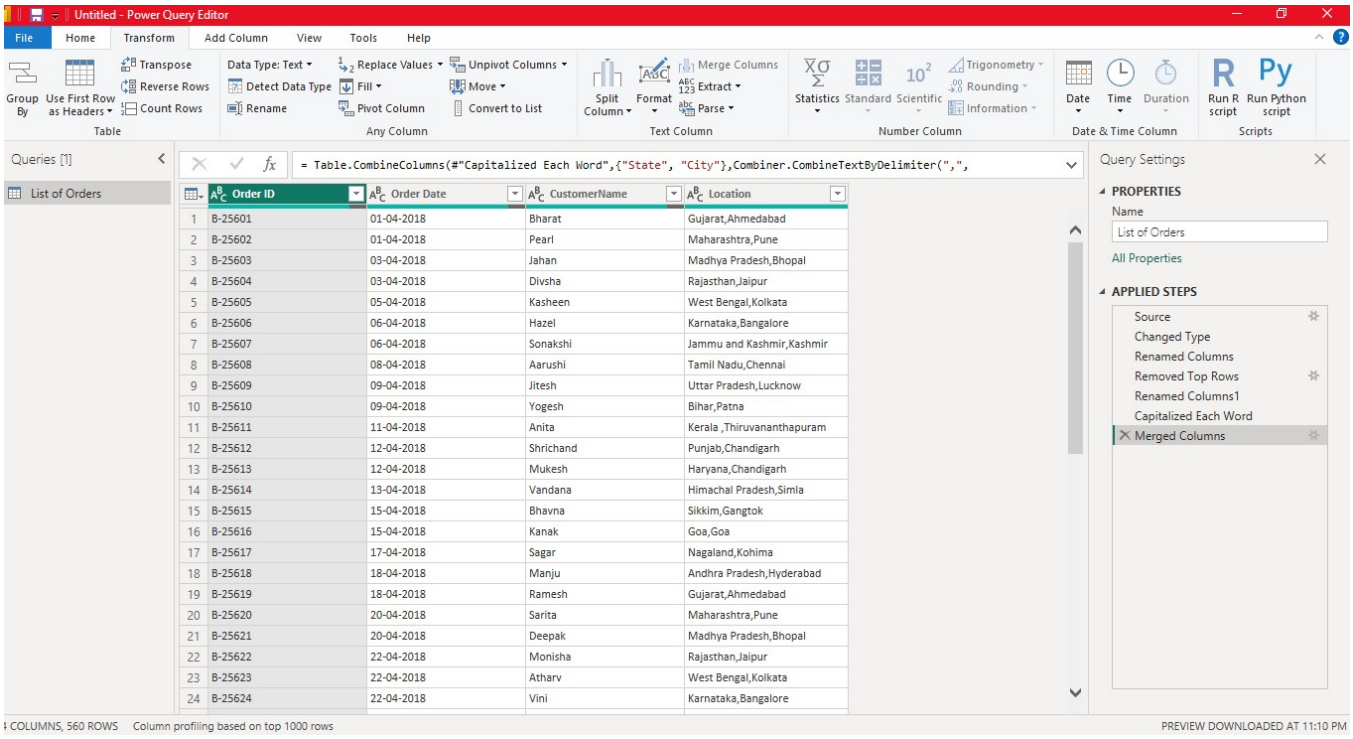
Screenshot 10



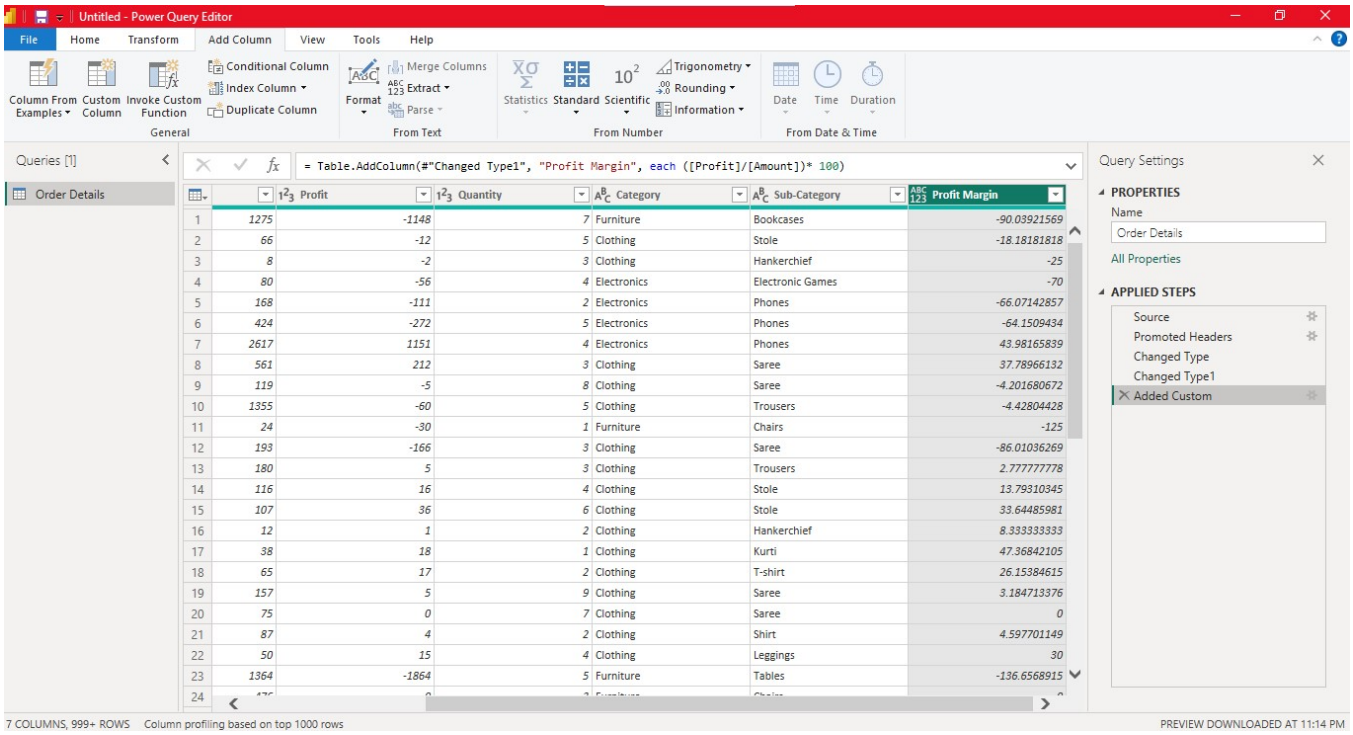
Screenshot 11



Screenshot 12

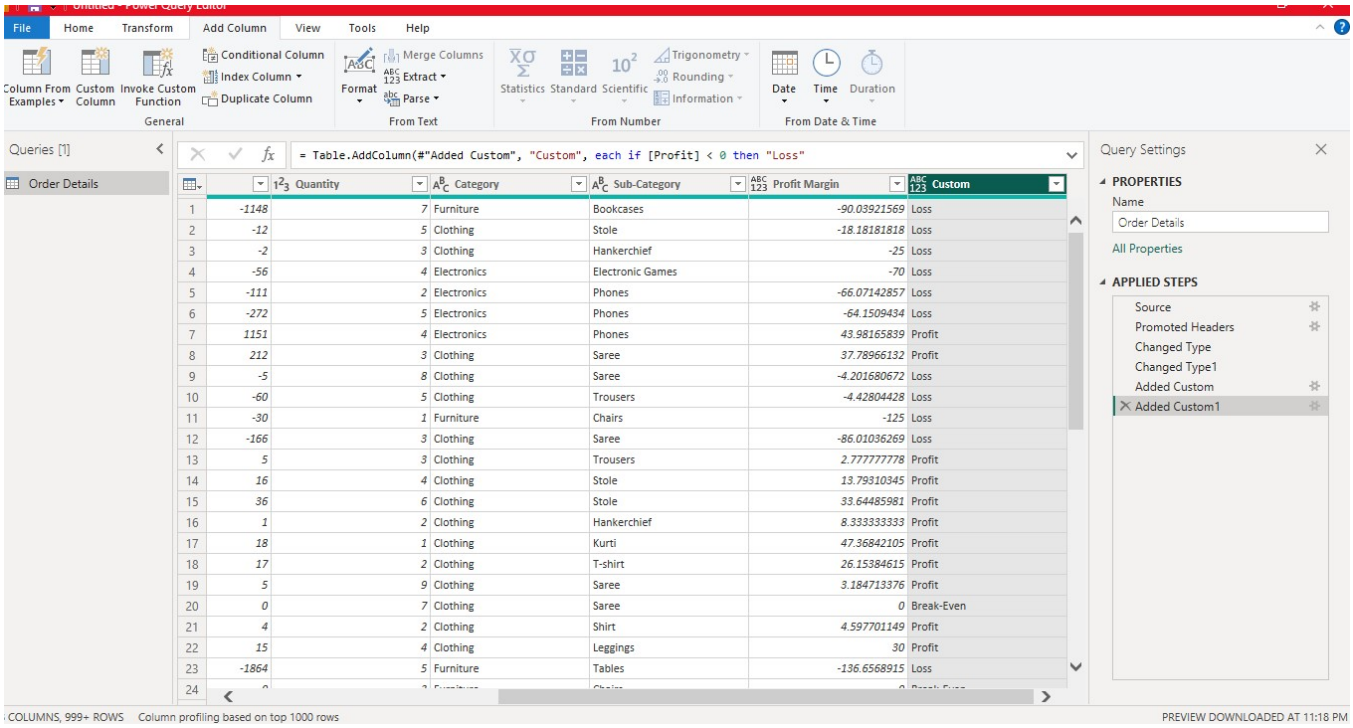


Screenshot 13



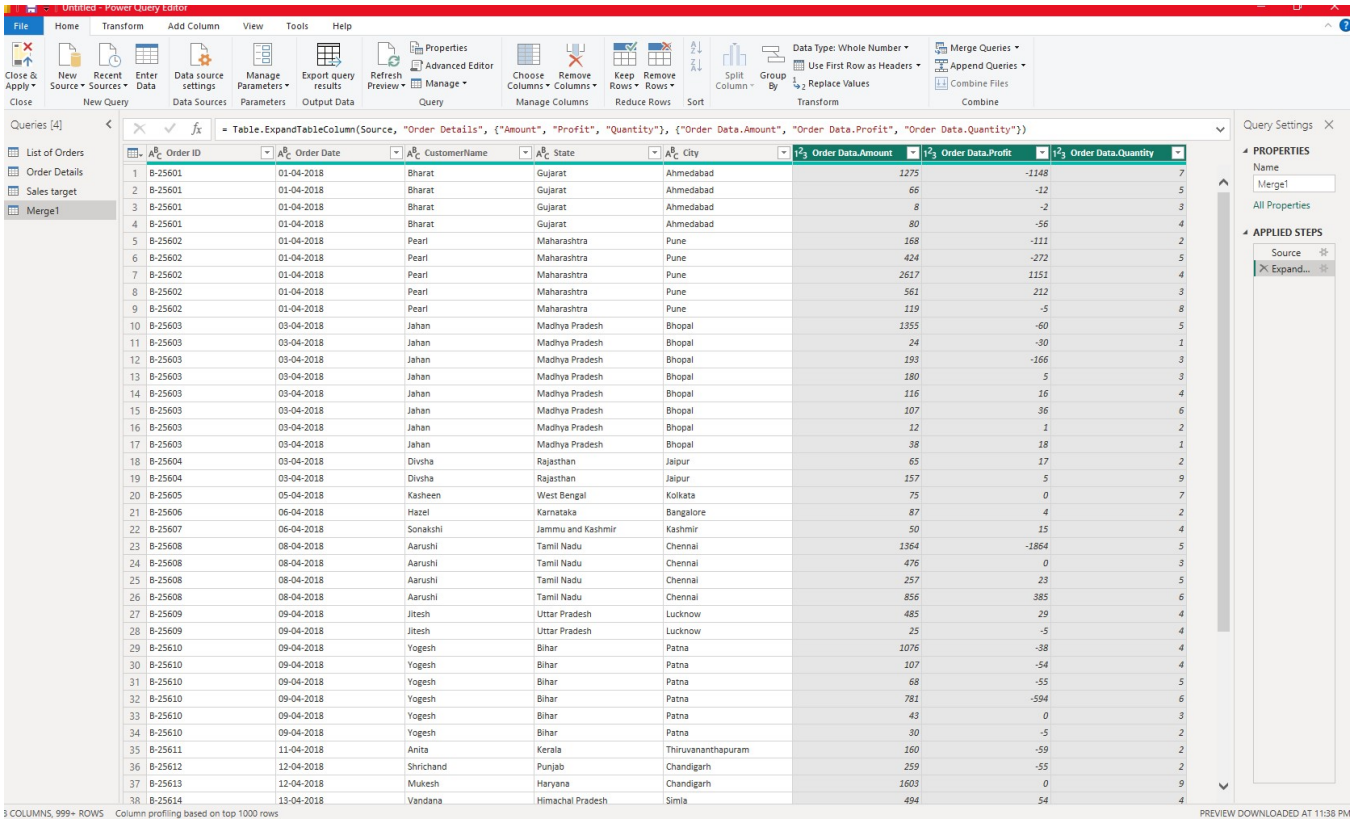
	Profit	Quantity	Category	Sub-Category	Profit Margin
1	1275	-1148	7 Furniture	Bookcases	-90.03921569
2	66	-12	5 Clothing	Stole	-18.18181818
3	8	-2	3 Clothing	Hankerchief	-25
4	80	-56	4 Electronics	Electronic Games	-70
5	168	-111	2 Electronics	Phones	-66.07142857
6	424	-272	5 Electronics	Phones	-64.1509434
7	2617	1151	4 Electronics	Phones	43.98165839
8	561	212	3 Clothing	Saree	37.78966132
9	119	-5	8 Clothing	Saree	-4.201680672
10	1355	-60	5 Clothing	Trousers	-4.42804428
11	24	-30	1 Furniture	Chairs	-125
12	193	-166	3 Clothing	Saree	-86.01036269
13	180	5	3 Clothing	Trousers	2.777777778
14	116	16	4 Clothing	Stole	13.79310345
15	107	36	6 Clothing	Stole	33.64485981
16	12	1	2 Clothing	Hankerchief	8.333333333
17	38	18	1 Clothing	Kurti	47.36842105
18	65	17	2 Clothing	T-shirt	26.15384615
19	157	5	9 Clothing	Saree	3.184713376
20	75	0	7 Clothing	Saree	0
21	87	4	2 Clothing	Shirt	4.597701149
22	50	15	4 Clothing	Leggings	30
23	1364	-1864	5 Furniture	Tables	-136.6568915

Screenshot 14

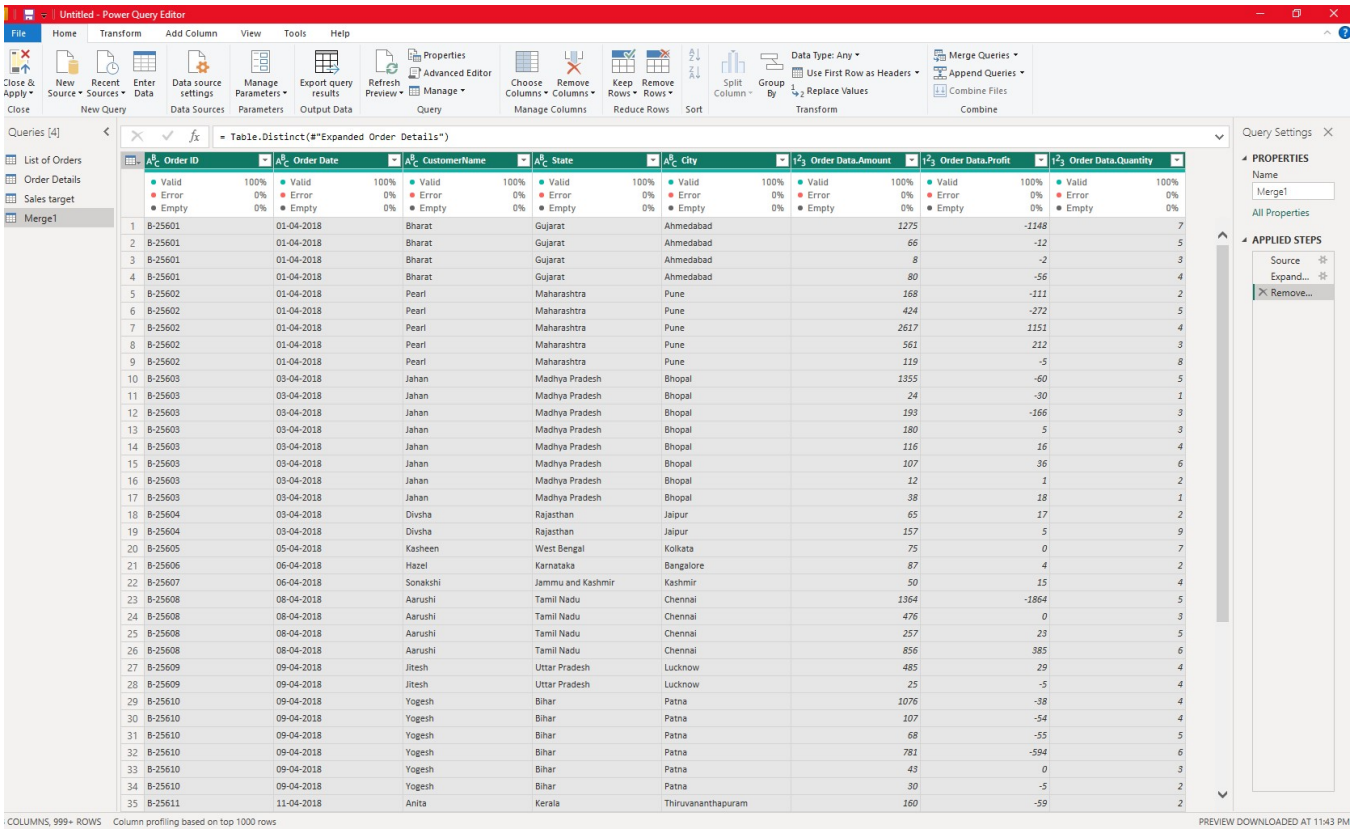


	Quantity	Category	Sub-Category	Profit Margin	Custom
1	-1148	7	Furniture	Bookcases	-90.03921569	Loss
2	-12	5	Clothing	Stole	-18.18181818	Loss
3	-2	3	Clothing	Hankerchief	-25	Loss
4	-56	4	Electronics	Electronic Games	-70	Loss
5	-111	2	Electronics	Phones	-66.07142857	Loss
6	-272	5	Electronics	Phones	-64.1509434	Loss
7	1151	4	Electronics	Phones	43.98165839	Profit
8	212	3	Clothing	Saree	37.78966132	Profit
9	-5	8	Clothing	Saree	-4.201680672	Loss
10	-60	5	Clothing	Trousers	-4.42804428	Loss
11	-30	1	Furniture	Chairs	-125	Loss
12	-166	3	Clothing	Saree	-86.01036269	Loss
13	5	3	Clothing	Trousers	2.777777778	Profit
14	16	4	Clothing	Stole	13.79310345	Profit
15	36	6	Clothing	Stole	33.64485981	Profit
16	1	2	Clothing	Hankerchief	8.333333333	Profit
17	18	1	Clothing	Kurti	47.36842105	Profit
18	17	2	Clothing	T-shirt	26.15384615	Profit
19	5	9	Clothing	Saree	3.184713376	Profit
20	0	7	Clothing	Saree	0	Break-Even
21	4	2	Clothing	Shirt	4.597701149	Profit
22	15	4	Clothing	Leggings	30	Profit
23	-1864	5	Furniture	Tables	-136.6568915	Loss
24						

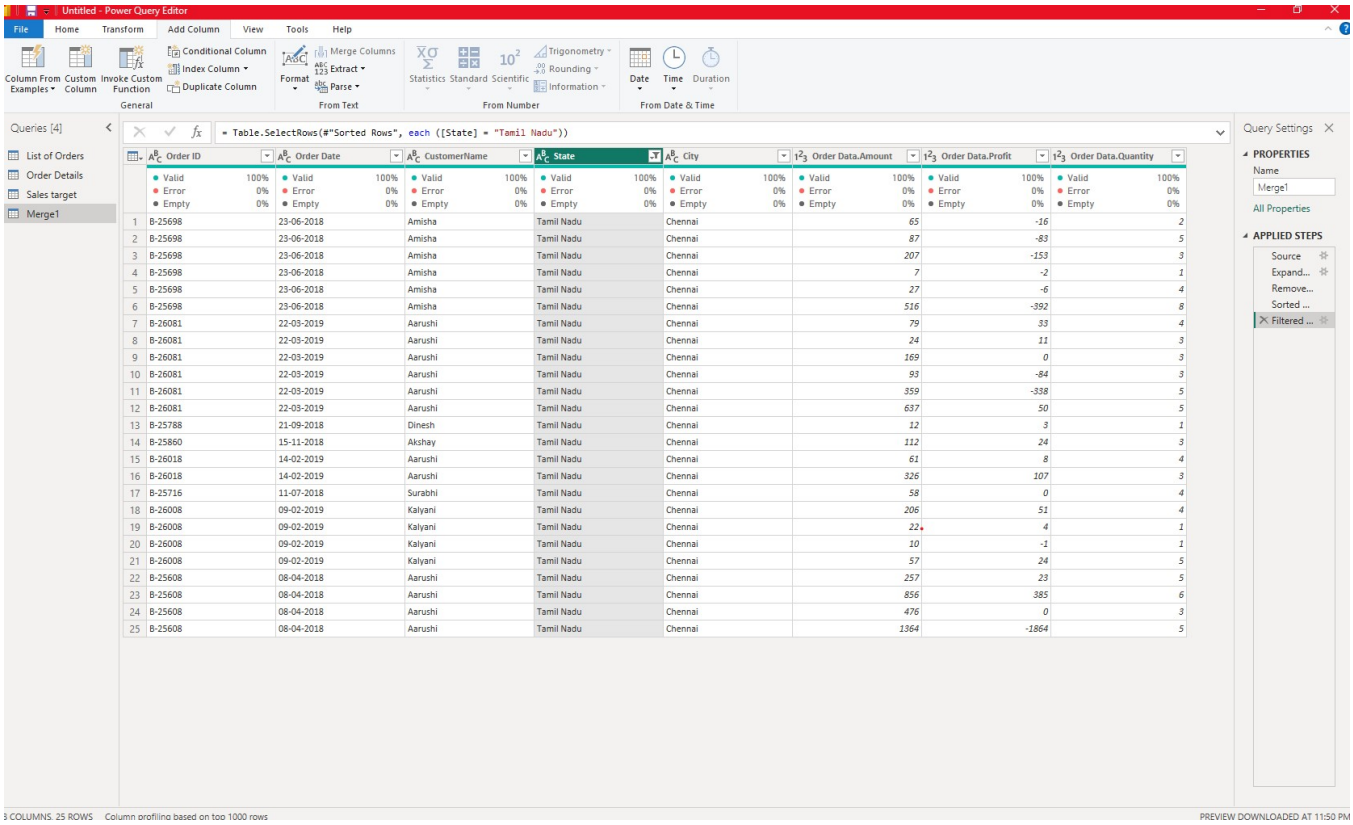
Screenshot 15



Screenshot 16



Screenshot 17



	Order ID	Order Date	CustomerName	State	City	Order Data.Amount	Order Data.Profit	Order Data.Quantity
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	B-25698	23-06-2018	Amisha	Tamil Nadu	Chennai	65	-16	2
2	B-25698	23-06-2018	Amisha	Tamil Nadu	Chennai	87	-83	5
3	B-25698	23-06-2018	Amisha	Tamil Nadu	Chennai	207	-153	3
4	B-25698	23-06-2018	Amisha	Tamil Nadu	Chennai	7	-2	1
5	B-25698	23-06-2018	Amisha	Tamil Nadu	Chennai	27	-6	4
6	B-25698	23-06-2018	Amisha	Tamil Nadu	Chennai	516	-392	8
7	B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	79	33	4
8	B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	24	11	3
9	B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	169	0	3
10	B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	93	-84	3
11	B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	359	-338	5
12	B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	637	50	5
13	B-25788	21-09-2018	Dinesh	Tamil Nadu	Chennai	12	3	1
14	B-25860	15-11-2018	Akshay	Tamil Nadu	Chennai	112	24	3
15	B-26018	14-02-2019	Aarushi	Tamil Nadu	Chennai	61	8	4
16	B-26018	14-02-2019	Aarushi	Tamil Nadu	Chennai	326	107	3
17	B-25716	11-07-2018	Surabhi	Tamil Nadu	Chennai	58	0	4
18	B-26008	09-02-2019	Kalyani	Tamil Nadu	Chennai	206	51	4
19	B-26008	09-02-2019	Kalyani	Tamil Nadu	Chennai	22	4	1
20	B-26008	09-02-2019	Kalyani	Tamil Nadu	Chennai	10	-1	1
21	B-26008	09-02-2019	Kalyani	Tamil Nadu	Chennai	57	24	5
22	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	257	23	5
23	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	856	385	6
24	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	476	0	3
25	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	1364	-1864	5

Screenshot 18

